

# XVAMP: Modeling the Effects of the Venus Atmosphere on X-Band Radar with Python

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## Software

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## Summary

One of the scientifically fascinating aspects about Venus is how it is very similar to Earth in some regards, while being completely dissimilar in others. For example, it is a rocky planet only marginally smaller than Earth, but it does not have a liquid water ocean (currently or potentially ever) and its toxic atmosphere is about 92 times as dense as ours. The upcoming NASA mission Venus Emissivity, Radio Science, InSAR, Topography, and Spectroscopy (VERITAS), currently scheduled for a launch in 2032, is a spacecraft designed to answer some of the key questions that could explain how a planet starting out similar to Earth in the habitable zone of our Sun could end up so differently ([Smrekar et al., 2022](#)). Among its instruments is the Venus Interferometric Synthetic Aperture Radar (VISAR), which will image the surface using X-band radar of 7.9 GHz using its two antennas. Apart from high-resolution, near-global surface imagery, one of VISAR's goals is to produce a Digital Elevation Model (DEM) two orders of magnitude better than existing datasets using its interferometric capabilities. However, the thick atmosphere strongly influences the radar signal, both attenuating the signal strength as well as bending the path of the electromagnetic waves (similar to optical waves bending as they pass between air and water). Without proper compensation, the DEM would be subject to errors on the order of tens to hundreds of meters, far above the VERITAS requirements. This study presents the X-Band Venus Atmosphere Model Permissivity (XVAMP) Python package, which implements the atmospheric model currently used by the VISAR engineering team. It is based on the atmospheric mixing model by Duan et al. (2010) and is the first publicly-available code for this topic to the knowledge of the authors. By making this software open-source, we solicit feedback from the scientific community already during the implementation phase of the mission, such that the model can truly represent the current state of our knowledge of the Venus atmosphere as it concerns radar.

## Statement of need

The VERITAS mission requires a model that for a given viewing geometry (angle and altitude) of the VISAR instrument onboard the spacecraft looking down at the planet yields the expected signal attenuation and path delay. The path delay is the difference between the direct geometric distance between the spacecraft and the surface and the actual distance traveled by the signal after accounting for the refraction in the atmosphere (which bends the path). On Venus, this delay can reach hundreds of meters; an error which would get directly passed onto the extracted DEM. The knowledge of the path delay is used onboard during the radar image formation process (which, for radar, requires the knowledge of the instrument velocity relative to the ground, which can be measured by the spacecraft but is corrupted by the atmosphere) and on the ground, where the path delay has to be removed when generating the DEM (which is derived from the phase difference between the images created simultaneously by the two VISAR antennas). Such an atmospheric model is strictly more than a model solely containing

the temperature, pressure, density, and species volume fraction at different altitudes such as the ones provided by Seiff et al. (1985) and Keating et al. (1985). This model is also different from General Circulation Models (GCM), which focus on the physical properties (e.g., wind speed) and include chemical reactions (in order to assess the lifetime of chemical species); e.g. Lebonnois et al. (2010). In particular, such a model is also more important for an X-band radar such as the one used by VERITAS compared to radar instruments of lower frequencies (such as the earlier Magellan mission by NASA, which produced the highest-resolution DEM of Venus to date, or the upcoming EnVision mission by ESA; both using S-band frequencies of 2.4 and 3.2 GHz, respectively) as the higher frequency is subject to larger attenuation.

## State of the field

Currently, there exists no publicly-available code to model the effect of the Venus atmosphere on radar signals to our knowledge. For Earth, the effect in X-band is much smaller, and there is less of a need to model the impact, since there are plenty of calibration methods available on Earth that are not possible on Venus (e.g., radar calibration sites using corner reflectors). The model used in the planning phase of the VERITAS mission is the proprietary, script-based code from Duan et al. (2010), which is difficult to integrate into other mission tools and to extend as newer datasets get published (or inferred by the VERITAS mission). Therefore, the present model is completely rewritten from scratch in order to become a state-of-the-art atmospheric model which will be used by the engineering team moving forward.

## Software design

The model of Duan et al. (2010) is based on empirical and analytical formulas which relate the effect of each chemical species in the atmosphere, scaled by their molar fraction and given the reigning temperature, pressure, and density profiles, to the effect on the *polarization*, *absorption*, and *refraction* of the Venus atmosphere. In addition, but in different manners, the size, density, and composition of the cloud layer as well as the electron density in the ionosphere are also incorporated. All of the aforementioned profiles are the key inputs to the Duan2010 model class, which is at the core of XVAMP. As a convenience for the user, many published profiles for the different constituents are included as datasets in the references module. During instantiation, all profiles are resampled to common altitude levels. Then, polarization parameters of the different species can either be loaded or computed (using data from laboratory measurements). Evaluating the polarization and absorption for each model constituent at each altitude level yields the complex permittivity profile, which can be converted to the refractive index. At this stage, the model provides integration routines through the absorption and refraction profiles along the curved path the radar wave takes, yielding the total delay and attenuation. Finally, geometric considerations in the geometry module yield the geometric range and angles as if no atmosphere was present, which are required for the data calibration. Throughout the processing steps of the model, units are preserved using the astropy package (The Astropy Collaboration et al., 2022) in order to ensure physical correctness and easier interpretation of the results.

## Research impact statement

The current impact of the presented model is its usage by the VISAR engineering team. In particular, this model is used to derive a polynomial approximation of the range delay and attenuation suitable to be stored in and evaluated by the data processor on board the spacecraft. It is also used by mission simulation tools during the design and implementation of the prototype DEM processing toolchain. Before and after arrival of the spacecraft at Venus, we furthermore welcome feedback from members of the wider scientific community to propose methodological improvements and incorporate new datasets the engineering team may not be

90 aware of. Finally, we expect end users of the scientific data to be released by VERITAS to use  
91 this model if they intend to reprocess the data themselves.

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## 97 AI usage disclosure

98 No generative AI tools were used in the development of this software, the writing of this  
99 manuscript, or the preparation of supporting materials.

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